

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's symptoms , including a sore throat , non-productive cough , fever , oropharyngeal erythema , and patchy reticular opacities on X-ray , are consistent with a respiratory infection . The description of oropharyngeal erythema and the presence of moist rales suggest a bacterial etiology . The patient was diagnosed with community-acquired pneumonia and improved with treatment with cephalexin , an antibiotic effective against Streptococcus pneumoniae and other bacteria . The immune response to Streptococcus pneumoniae primarily involves the activation of Th1 cells and the release of cytokines like TNF - Î± and IFN - Î³ , which are characteristic of a Th1 response . Peptidoglycan is not a major antigen of Streptococcus pneumoniae ; instead , it has a cell wall component called polysaccharide capsule , which can be recognized by TLR4 and can induce a Th1 response . Encapsulation of the pathogen is a mechanism of evading the immune response , but it is not the primary immune response elicited . TLR5 is involved in recognizing flagellin in other pathogens , not Streptococcus pneumoniae .

</think>

<answer> B . It primarily induces the Th1-cell response . </answer>